Continuously Varying Heart Rate in a Pulsatile Circulation Model with Feedback Control

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Abstract

When blood vessels dilate, the heart must supply more flow if arterial pressure is to be maintained. A pulsatile circulation model with a fixed heart rate cannot make this adjustment on its own. We introduce a continuously varying cardiac clock whose speed responds to filtered arterial pressure, allowing the interval between beats to change while preserving the sequence of ventricular contraction and relaxation. Simulations of an exercise-like reduction in systemic resistance show how heart-rate feedback increases flow and limits the fall in pressure. Raising the pressure target produces a further increase in both pressure and flow, but also changes the recovery after exercise. We explain why proportional feedback leaves a pressure shortfall whenever the required mean heart rate rises, why the rate response must remain bounded, and how delayed responses can lead to sustained oscillations when feedback is too strong. The model connects these behaviors to a small set of physical mechanisms and provides a simple setting for studying the interaction between pulsatile pumping and pressure regulation.

1 Introduction

During exercise, dilation of the vessels supplying active muscle reduces the resistance to blood flow. If the heart continued pumping at its resting rate, this reduction would tend to lower arterial pressure. The circulation responds through several interacting mechanisms, including changes in heart rate, contractility, vascular tone, and venous return. Here we isolate one part of that response: how changing the rate of a pulsatile pump helps regulate arterial pressure.

The starting point is a left ventricle that fills and ejects blood into a compliant arterial compartment. Contraction is represented by a periodic change in ventricular compliance. At a fixed heart rate, the compliance follows the same cycle at regular intervals. To allow heart rate to change, we introduce a cardiac clock: the ventricle still follows the same sequence of contraction and relaxation, but the clock can run faster or slower. A pressure-feedback rule adjusts its speed when the measured pressure differs from a prescribed target.

Baroreflex control has been studied in both averaged and pulsatile circulation models. Ottesen analyzed the role of time delay in a Windkessel feedback model [1], and Ursino coupled pulsatile heart dynamics to multiple baroreflex pathways and responses [2]. Other studies have examined regulation during hemorrhage [3] and the sensitivity of more detailed regulated circulation models to their parameters [4]. Our aim is to make the behavior of a much smaller model easy to follow. We ask what heart-rate feedback alone can accomplish, what remains outside its control, and why a response that helps at moderate strength can become oscillatory when made too strong.

The exercise experiment also distinguishes feedback from a change in the pressure target. Experimental evidence shows that baroreflex operating conditions can shift during exercise [5]. We represent that possibility by prescribing a higher target while resistance is reduced. This gives a way to compare two different actions: correcting a pressure drop around the original target, and

asking the circulation to operate at a higher pressure. The prescribed changes are an exercise-like experiment, rather than a complete physiological model of exercise.

2 A compartmental picture of the circulation

2.1 The path followed by the blood

Figure 1 shows the model. Blood enters the left ventricle from a reservoir held at a fixed left-atrial pressure. The mitral valve permits filling when atrial pressure exceeds ventricular pressure. As the ventricle contracts, its pressure rises; the aortic valve then permits ejection into the systemic arteries. Blood leaves the arterial compartment through the systemic resistance.

Only the ventricle and systemic arteries store variable amounts of blood in this model. The inlet and outlet are prescribed reservoirs. This keeps the model small enough to understand, but it also means that venous return and the total volume of a closed circulation are not represented. The *feedback loop* is closed because arterial pressure affects heart rate, even though the blood-flow model has open boundaries.

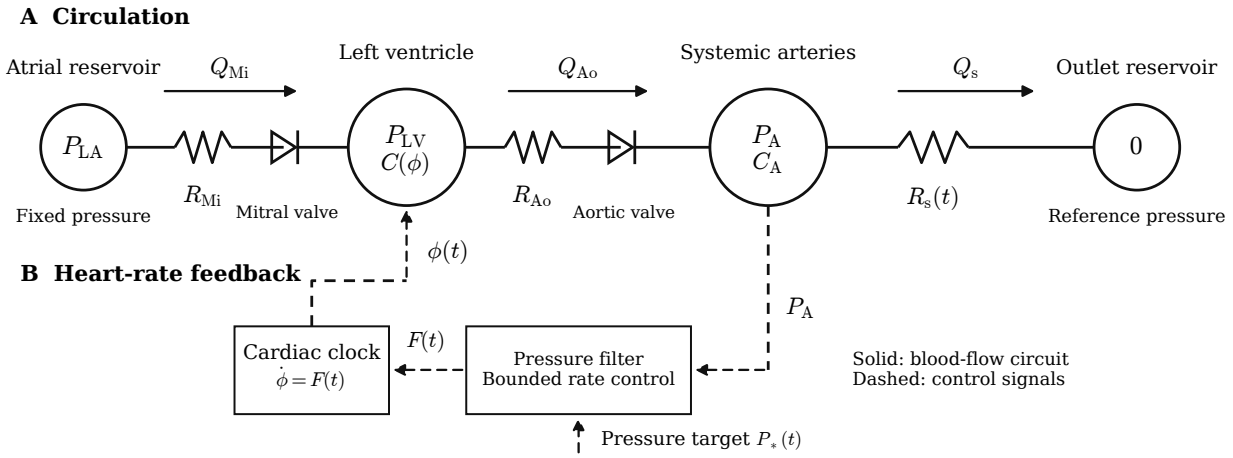


Figure 1: Circuit representation of the circulation and its controller. **A:** Circular chambers represent the left ventricle, with time-varying compliance $C(\phi)$, and systemic arteries, with constant compliance C_{A}. The atrial and outlet reservoirs have prescribed pressures. Zigzag elements represent the mitral, aortic, and systemic resistances; diode symbols enforce one-way flow through the valves. Arrows indicate positive blood flow. **B:** Dashed connections carry control signals: arterial pressure is filtered and compared with a target, the bounded rate response sets heart rate, and accumulated cardiac phase determines ventricular compliance. The reservoir boundaries remain open; the feedback path is not a blood-flow return path.

Time is measured in minutes, volume in litres, and pressure in mmHg. Let P_{LV} and P_{A} be ventricular and arterial pressure. We write the volume above the unstressed volume in each compartment as

$$x_{\mathrm{LV}} = C(\phi)P_{\mathrm{LV}}, \qquad x_{\mathrm{A}} = C_{\mathrm{A}}P_{\mathrm{A}}. \tag{1}$$

Here C_{A} is constant, while ventricular compliance C varies with cardiac phase ϕ. Compliance measures how much volume can be stored for a given pressure: a relaxed ventricle has high

compliance, whereas a contracting ventricle has low compliance. This description is the reciprocal of a time-varying elastance model [6].

For each compartment, the rate of volume change is simply inflow minus outflow:

$$\dot{x}_{\mathrm{LV}} = Q_{\mathrm{Mi}} - Q_{\mathrm{Ao}}, \tag{2}$$

$$\dot{x}_{\mathrm{A}} = Q_{\mathrm{Ao}} - Q_{\mathrm{s}}. \tag{3}$$

The valve flows depend on pressure differences, but they cannot reverse:

$$Q_{\mathrm{Mi}} = \frac{[P_{\mathrm{LA}} - P_{\mathrm{LV}}]_+}{R_{\mathrm{Mi}}}, \qquad Q_{\mathrm{Ao}} = \frac{[P_{\mathrm{LV}} - P_{\mathrm{A}}]_+}{R_{\mathrm{Ao}}}, \qquad Q_{\mathrm{s}} = \frac{P_{\mathrm{A}}}{R_{\mathrm{s}}}. \tag{4}$$

The notation $[z]_+ = \max(z, 0)$ means that a negative driving pressure closes a valve instead of producing backward flow. The outlet pressure is set to zero as the reference pressure.

Because ventricular compliance changes during a beat, its storage term is $d(CP_{\mathrm{LV}})/dt$. In particular,

$$C(\phi)\dot{P}_{\mathrm{LV}} + C'(\phi)\dot{\phi}P_{\mathrm{LV}} = Q_{\mathrm{Mi}} - Q_{\mathrm{Ao}}. \tag{5}$$

The second term expresses an important physical effect: ventricular pressure can rise when compliance falls even if the volume has not yet changed. Omitting it would remove the mechanism by which contraction generates pressure.

2.2 Letting the cardiac clock run faster or slower

We measure cardiac phase in cycles. Each increase of one in ϕ represents one heartbeat. If $F(t)$ is heart rate in beats per minute, then

$$\dot{\phi} = F(t), \qquad \phi(t) = \phi(0) + \int_0^t F(s)\, ds. \tag{6}$$

A larger rate advances the clock faster and shortens the time to the next beat. The integral matters: replacing phase by $F(t)t$ would introduce an extra term $t\dot{F}$ into the rate of phase change. For example, a rate that increases linearly would then accelerate the clock twice as quickly as intended.

Within each beat, let $\theta = \phi - \lfloor\phi\rfloor$ be the fraction of the cycle already completed. The compliance falls from its relaxed value C_{D} to its contracted value C_{S} during systole and returns during diastole:

$$C(\theta) = \begin{cases} C_{\mathrm{D}}(C_{\mathrm{S}}/C_{\mathrm{D}})^{e_{\mathrm{s}}(\theta)}, & 0 \le \theta < \sigma, \\ C_{\mathrm{S}}(C_{\mathrm{D}}/C_{\mathrm{S}})^{e_{\mathrm{d}}(\theta)}, & \sigma \le \theta < 1, \end{cases} \tag{7}$$

where

$$e_{\mathrm{s}} = \frac{1 - e^{-\theta/\eta_{\mathrm{s}}}}{1 - e^{-\sigma/\eta_{\mathrm{s}}}}, \qquad e_{\mathrm{d}} = \frac{1 - e^{-(\theta-\sigma)/\eta_{\mathrm{d}}}}{1 - e^{-(1-\sigma)/\eta_{\mathrm{d}}}}. \tag{8}$$

The normalized exponential factors make the compliance reach the same two endpoints on every cycle. Heart-rate control changes *when* these values are reached, rather than changing the values themselves. Both systole and diastole are compressed when the clock runs faster. Separate changes in contractility or relaxation are not included.

3 How pressure controls heart rate

3.1 Responding to a trend rather than every pressure pulse

Arterial pressure rises and falls within each heartbeat. Feeding those rapid changes directly into the rate rule would make the controller respond to the pulse it has just helped create. We therefore use a filtered pressure p_f:

$$\tau_{\mathrm{P}} \dot{p}_f = P_{\mathrm{A}} - p_f. \tag{9}$$

The filter continually moves toward the current arterial pressure. A longer time constant averages over more of the beat-to-beat variation, but also makes the pressure estimate slower to respond to a genuine change.

We compare this filtered pressure with a target P_* and prescribe a rate command

$$u = F_0 \left[1 + g \frac{P_* - p_f}{P_*} \right], \qquad g \geq 0. \tag{10}$$

At the target pressure, the command is the reference rate F_0. Below the target, the command increases; above it, the command decreases. The dimensionless gain g determines how strongly the heart rate responds to the same relative pressure error. This rule is a simple negative-feedback representation of pressure regulation, without explicit neural pathways.

3.2 Keeping the rate within bounds

Equation (10) is linear in the pressure error. Taken literally, it can request arbitrarily high or even negative rates. A negative rate would run the cardiac clock backward, which has no intended physiological meaning. We therefore limit the command to an interval:

$$u_b = \begin{cases} F_{\min}, & u < F_{\min}, \\ u, & F_{\min} \leq u \leq F_{\max}, \\ F_{\max}, & u > F_{\max}. \end{cases} \tag{11}$$

This operation is often called *clipping* or *saturation*. It leaves commands inside the interval unchanged and replaces commands outside it by the nearest bound. In these experiments, the interval is 30–240 bpm. These are illustrative model choices, not fitted physiological limits.

In the simplest version, the actual rate changes immediately to $F = u_b$. To study a gradual response, we also consider

$$\tau_{\mathrm{H}} \dot{F} = u_b - F. \tag{12}$$

Now the heart rate moves toward the command instead of jumping to it. The time constant τ_{H} sets the speed of this adjustment. This is a response lag, not a fixed waiting time before the response begins. With a positive lower bound and an initial rate inside the interval, the actual rate stays inside the same bounds and cardiac phase always advances.

A sudden downward change in the pressure target illustrates why the bounds matter. Immediately after the change, the filtered pressure still reflects the preceding high-pressure state. The controller may therefore request a very large reduction in rate. For $g > 0$, the unrestricted command remains positive only when

$$p_f < P_* \left(1 + \frac{1}{g} \right). \tag{13}$$

Increasing gain narrows the range of pressures for which an unrestricted command is positive. Bounding the command makes the model well defined even during such a target reset.

4 An exercise-like experiment

We first let the model settle under resting conditions. At 30 s, systemic resistance is reduced from 17.86 to 9 mmHg min/L, representing vasodilation at the start of exercise. At 60 s, resistance is restored. The simulation continues to 90 s so that recovery can be observed. The shaded interval in Fig. 2 marks exercise; the start and stop are indicated by dashed lines.

We compare four cases. In the first, the heart remains at 80 bpm. In the second, pressure feedback can change the rate but the pressure target stays at its resting value. In the third, the target rises to 150 mmHg during exercise and returns to its resting value at exercise stop. The fourth uses the same changing target but adds a gradual heart-rate response. This order lets us separate the roles of feedback, target resetting, and response lag.

Parameter	Value	Unit
Reference period T_0; rate F_0	0.0125; 80	min; bpm
Systolic fraction σ	0.4	cycle fraction
Shape constants η_s, η_d	0.2, 0.6	cycle fractions
Ventricular C_S, C_D	0.00003, 0.0146	L/mmHg
Arterial C_A	0.00175	L/mmHg
Valve resistances R_{Mi}, R_{Ao}	0.01, 0.01	mmHg min/L
Systemic R_s: rest; exercise	17.86; 9	mmHg min/L
Inlet pressure P_{LA}	5	mmHg
Target P_*: rest; raised target	93.4709863908; 150	mmHg
Pressure-filter time τ_P	3	s
Optional rate-response time τ_H	0.6	s
Feedback gain g	10	dimensionless

Table 1: Parameters for the main experiment. Times shown in seconds are converted to minutes in the equations and implementation.

The initial ventricular and arterial pressures are 5 and 80 mmHg, respectively, with the filter initialized at the resting target, heart rate at 80 bpm, and phase at zero. The 30 s before exercise allows the initial adjustment to decay. We report mean pressure, flow, and rate over the last nine seconds of exercise. These are averages over the same time interval in every case, not estimates based on the height of individual pressure peaks.

The equations are solved in a form that balances the blood entering and leaving each compartment. A time step of 1.875 ms is used for the main plots; smaller steps and a separate adaptive solver check the results. The advanced interpretation in Appendix A develops the mathematical analysis, with numerical checks in Section A.9.

5 What the model shows

5.1 A faster heart helps maintain pressure when resistance falls

With heart rate fixed, arterial pressure falls markedly when the systemic vessels dilate. Blood can leave the arteries more easily, but the pump has no control mechanism that tells it to beat faster. Mean flow rises only slightly. Pressure feedback changes this outcome: the pressure drop increases the rate command, the cardiac clock advances faster, and more blood is delivered per minute.

Table 2 shows the size of the effect. At fixed rate, mean pressure falls to about 48 mmHg. With feedback and a fixed pressure target, the mean heart rate rises to about 153 bpm, flow nearly

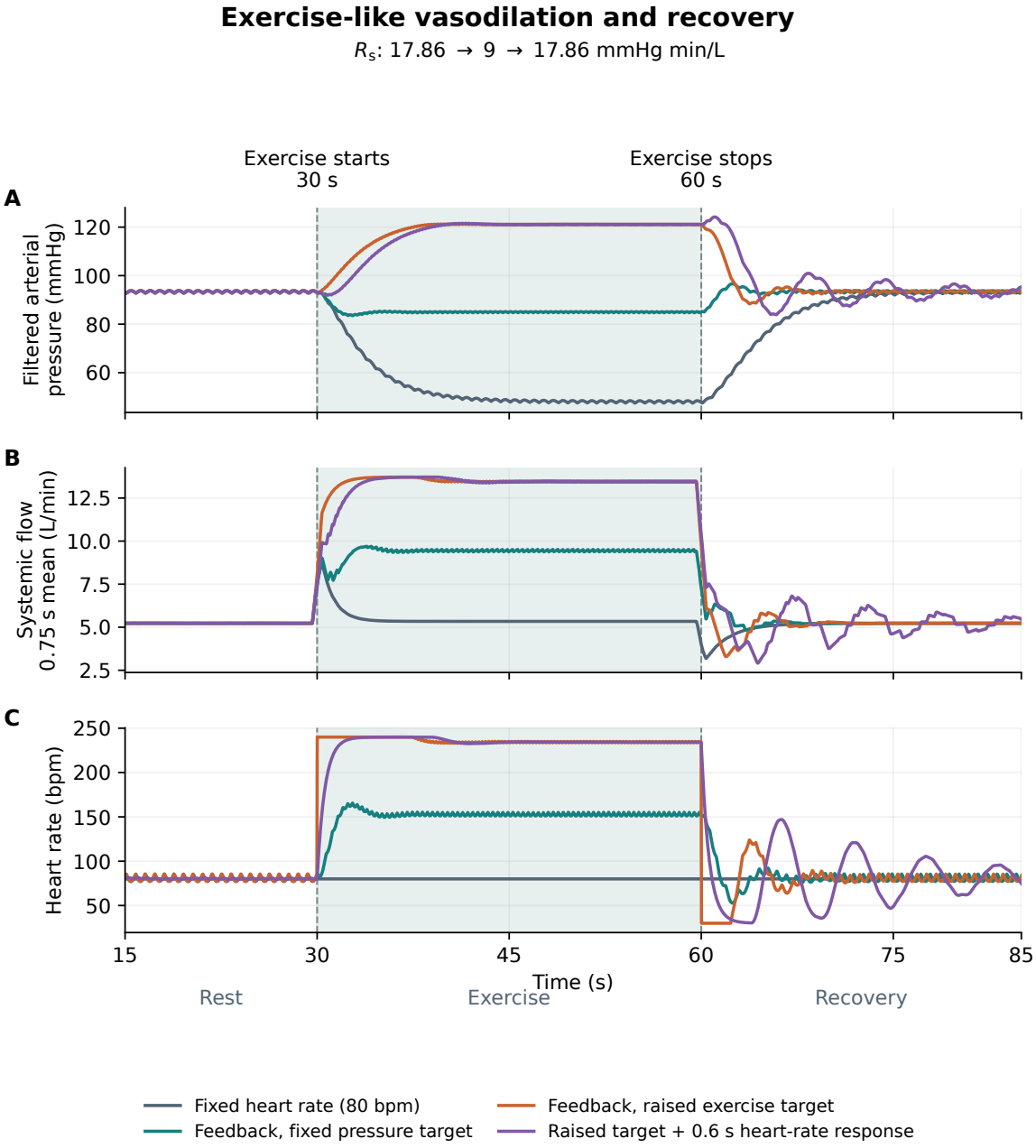


Figure 2: Pressure, flow, and heart rate during exercise-like vasodilation. Resistance falls at *exercise start* (30 s) and returns at *exercise stop* (60 s). The pressure target stays at its resting value for the fixed-rate and fixed-target cases; it rises during exercise in the two raised-target cases. **A:** Filtered arterial pressure. **B:** Systemic flow, displayed with a centered 0.75 s moving average. **C:** Heart rate. Feedback gain is 10 and the pressure-filter time is 3 s.

Controller	Mean pressure mmHg	Mean flow L/min	Mean rate bpm
Fixed rate	48.06	5.340	80.00
Fixed target	84.98	9.443	152.58
Raised target	121.06	13.451	234.20
Raised target + rate response	121.12	13.458	234.17

Table 2: Averages over 51–60 s, near the end of exercise. Controlled cases use rate bounds of 30–240 bpm.

doubles relative to rest, and mean pressure remains near 85 mmHg. The feedback therefore opposes the pressure drop, although it does not fully remove it. The next subsection explains why this remaining shortfall is a consequence of the control rule itself.

5.2 Why a fixed target still leaves a pressure shortfall

The controller in Eq. (10) needs a pressure error to sustain a rate different from F_0. If pressure returned exactly to the target, the command would return to F_0 as well. A circulation that needs a higher mean rate to supply the required flow must therefore remain, on average, below the pressure target.

This argument can be made exact for a periodic response with constant target and inactive rate bounds. Denote a time average over a period by an overbar. Averaging Eq. (9) gives $\overline{p_f} = \overline{P_{\mathrm{A}}}$ because the filter ends the period where it started. A gradual rate response similarly gives $\overline{F} = \overline{u}$. Averaging the command then yields

$$\boxed{\frac{P_* - \overline{P_{\mathrm{A}}}}{P_*} = \frac{\overline{F}/F_0 - 1}{g}.} \tag{14}$$

The left side is the relative pressure shortfall. The right side is the relative increase in mean heart rate, divided by the feedback gain. This relation does not assume a fixed stroke volume, and it continues to hold with a gradual unsaturated rate response.

For the fixed-target exercise experiment, the observed increase in heart rate corresponds to a pressure shortfall of about 8.5 mmHg, close to the simulation average. Small differences can arise because the reported window need not contain an exact number of periods. More generally, endpoint changes in the filter and rate must be included when averaging a transient; the correction is given in Section A.3.1.

The volume balances provide another useful relation. On a periodic response at constant resistance, neither compartment gains volume from one period to the next. Their mean inflows and outflows must therefore agree:

$$\overline{Q_{\mathrm{Mi}}} = \overline{Q_{\mathrm{Ao}}} = \overline{Q_{\mathrm{s}}} = \frac{\overline{P_{\mathrm{A}}}}{R_{\mathrm{s}}}. \tag{15}$$

This explains why maintaining pressure at a lower resistance requires a larger mean flow. It is the additional flow requirement that makes a sustained increase in rate necessary in this model.

5.3 A higher target asks the circulation to do more

Raising the target while resistance is reduced makes the controller request a larger increase in rate. Both pressure and flow then rise above their resting values. In the main experiment, mean flow

reaches about 13.45 L/min and the mean rate approaches the upper bound. Adding a gradual rate response has little effect on the late-exercise averages, but it changes how the system reaches them and how it recovers afterward.

The higher pressure target is an input to the model. It represents a changed operating condition; it is not a change that this feedback rule predicts by itself. This distinction is useful when interpreting the experiment. Feedback around the resting target tries to reduce a pressure loss, whereas a higher target asks for a pressure increase despite the lower resistance.

At exercise stop, the target returns to its resting value while the pressure filter still remembers the preceding high-pressure state. Heart rate consequently falls sharply. With unrestricted control, this mismatch can make the requested rate negative, as anticipated by Eq. (13). The bounded rule keeps the clock moving forward. A gradual rate response smooths the immediate change but adds memory, which can make recovery oscillatory.

5.4 Why increasing rate does not preserve the volume ejected per beat

A useful way to separate pump behavior from the controller is to prescribe a series of constant heart rates. For each rate, we allow the circulation to settle and then measure mean pressure and the mean volume ejected per beat. Figure 3 shows these operating curves at resting and exercise resistance.

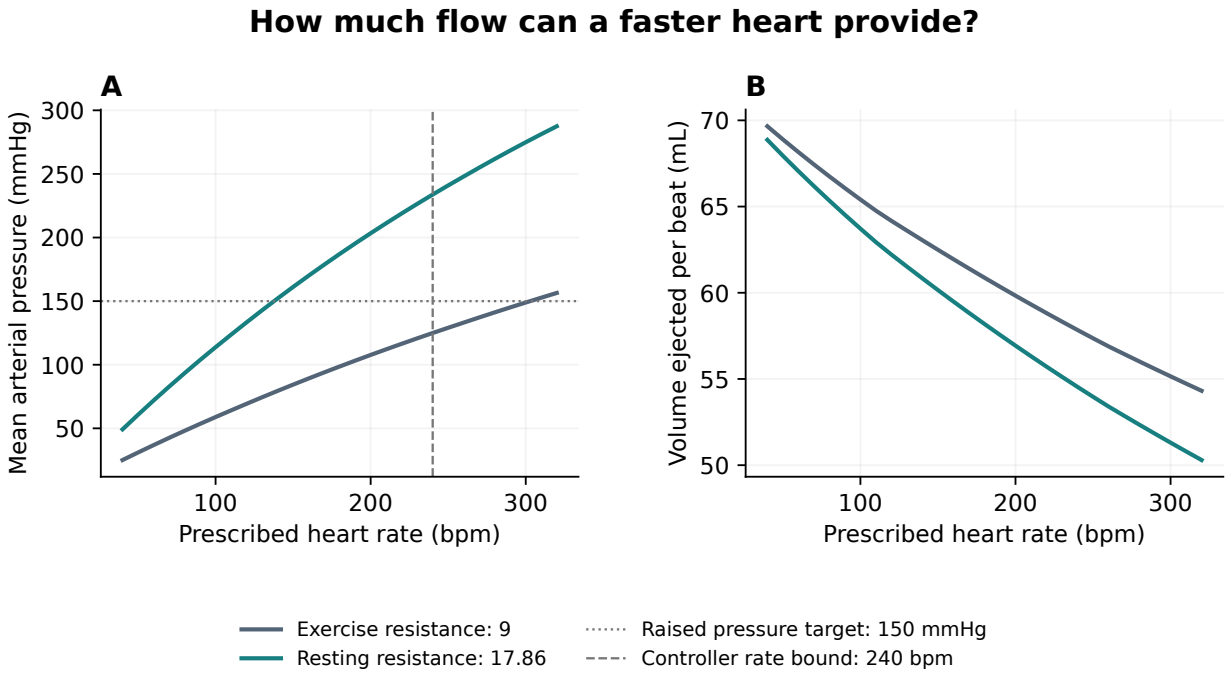


Figure 3: Fixed-rate experiments after the circulation has settled. **A:** Mean pressure as the prescribed rate is increased. **B:** Mean ejected volume per beat, calculated as mean outflow divided by heart rate. Each curve uses a fixed resistance, given in mmHg min/L. The horizontal reference marks the raised pressure target; the vertical reference marks the upper rate bound used in the feedback experiments.

The volume ejected per beat falls as rate rises. Running the cardiac clock faster shortens both

filling and ejection, so multiplying the resting stroke volume by an arbitrarily high rate would overestimate output. With exercise resistance and a fixed rate of 240 bpm, the mean pressure is about 125 mmHg, below the raised target of 150 mmHg. The controller therefore has limited room to improve pressure by rate changes alone.

These curves describe this particular ventricle, compliance waveform, and atrial reservoir. They are not a universal limit on the circulation, nor a proof of the largest pressure possible under every time-varying control. Changing filling conditions or contractility could change the amount of blood delivered on each beat.

6 Feedback strength, averaging, and response time

6.1 Stronger feedback responds to smaller errors

Equation (14) suggests one way to reduce the pressure shortfall: increase the gain. With stronger feedback, a smaller pressure error can support the required rate increase. However, the controller also becomes more responsive to pressure variation that passes through the filter. A short filter time and high gain can therefore cause large within-beat changes in the rate command.

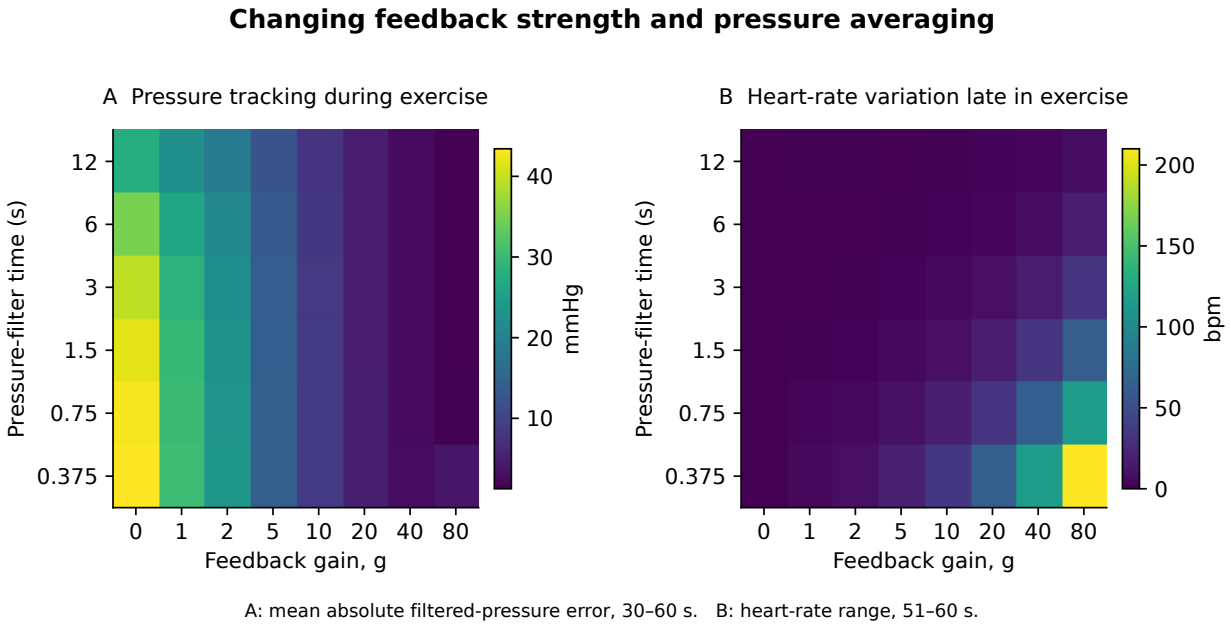


Figure 4: The same exercise experiment repeated at different gains and filter times, with a fixed pressure target and instantaneous bounded rate response. **A:** Mean absolute difference between filtered pressure and the target during exercise (30–60 s). **B:** Range of heart rate near the end of exercise (51–60 s). The right panel includes variation within each heartbeat; it does not, by itself, measure long-term instability.

Figure 4 shows this tradeoff. At high gain with a short averaging time, the rate can move across the full allowed interval and spend much of the exercise period at a bound. A longer filter time suppresses more of the pulse, but it also delays recognition of a sustained pressure change. In particular, a small error in the *filtered* signal should not be mistaken for a small error in the actual

pressure during a rapid transient. There is no single best gain or filter time independent of the response one wants to control.

6.2 Why a delayed response can overcorrect

The pressure filter already introduces memory: it reflects recent pressure, not just the present value. A gradual rate response introduces another memory. When pressure starts to recover, the command may still be asking the heart to speed up, and the actual rate may still be catching up to that command. The circulation can then overshoot. If the response is sufficiently strong, successive corrections need not die away.

Averaging over the individual pressure pulses gives a simpler description of this effect. In that approximation, pressure rises when mean pump inflow exceeds arterial outflow, the pressure filter follows that change, and the heart-rate response follows the filtered error. These three coupled responses can reinforce an overshoot. The linearized equations and their stability criterion are developed in Section A.6.

Without a rate-response lag, the approximation predicts that small disturbances decay for the restoring pump response considered here. Adding the lag changes the result: decay now requires the gain to remain below a threshold. The algebraic condition is derived in Section A.6. With the parameters used here, it places the threshold near gain 14.2. This gives a prediction to check in the full model, where the heart still beats and the valves still open and close.

6.3 Following a small disturbance from one beat to the next

We test the full model by comparing its state at the same point in successive heartbeats. Begin with a repeating beat, perturb its pressures and rate slightly, and ask whether that difference gets smaller or larger after the next beat. Repeating this comparison removes the distraction of comparing different phases of the cardiac cycle.

Mathematically, the comparison defines a one-beat return map. The largest magnitude of its linearized eigenvalues, often called a Floquet multiplier, measures the growth or decay of small disturbances over repeated beats. A value below one indicates decay; a value above one indicates growth in at least one direction. Figure 5 uses this measure to show how stability changes with feedback gain.

The repeating beat is stable at gain 14 and unstable at gain 16. An independent adaptive solver gives the same classification. The rate bounds are inactive on the repeating beats used for this local calculation, so this transition is not caused by the controller hitting a bound. Without the added rate-response lag, the tested repeating beats at gains 10 and 40 remain stable. The response time therefore changes the consequences of increasing gain.

Longer simulations show what can happen after the repeating beat loses stability (Fig. 6). At moderate gain, the late pressure signal has only a small variation. At high gain, large oscillations persist and the rate command frequently reaches a bound. The local stability calculation explains why small disturbances no longer decay; the time traces show how the bounded controller behaves once those disturbances become large. The simulations do not by themselves identify a unique nonlinear oscillation or prove a particular bifurcation type.

7 Filling conditions and recovery also matter

Heart rate is not the only factor controlling how much blood the ventricle can deliver. To examine this point, we change individual model parameters while keeping the feedback rule and pressure

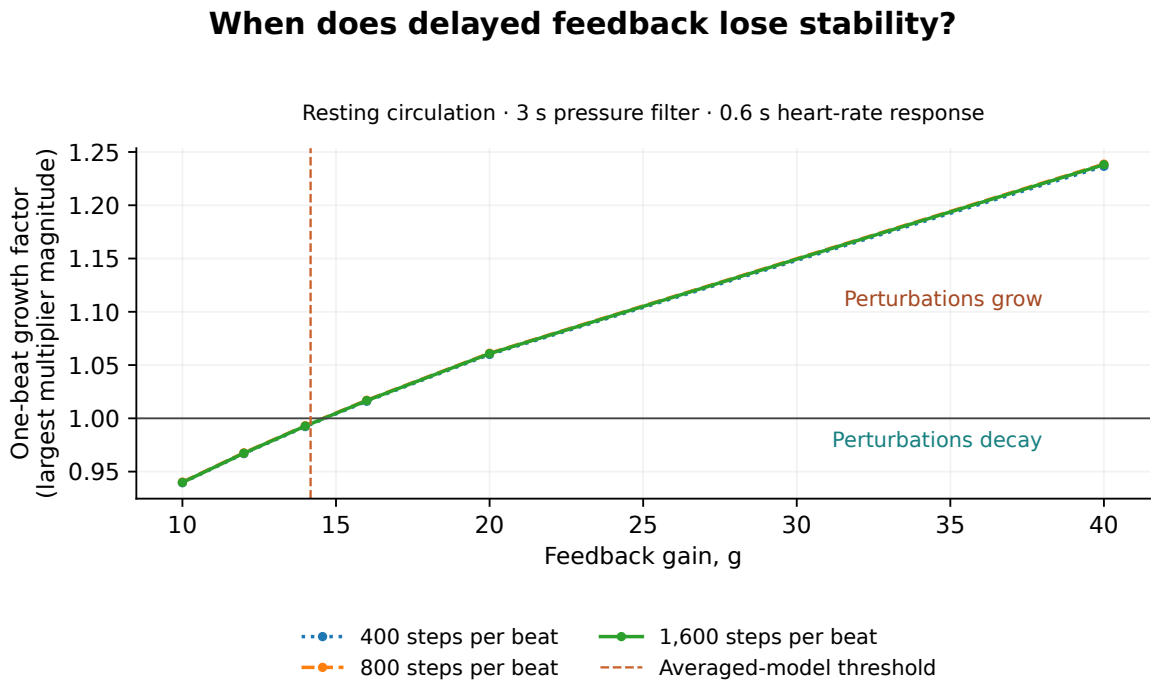


Figure 5: Stability of the resting pulsatile response with a 3 s pressure filter and a 0.6 s rate-response time. The vertical axis is the largest magnitude of the one-beat return-map multipliers. Values below one correspond to decaying disturbances; values above one indicate instability of the repeating beat. Three step sizes give closely agreeing curves. The averaged-model threshold is shown for comparison.

target fixed. Lowering atrial pressure by 20% reduces the pressure available for filling. During exercise the controller then raises mean rate to about 188 bpm, yet mean pressure remains lower than in the baseline case. Lowering relaxed ventricular compliance has a similar effect because it reduces the volume stored at a given filling pressure. The full comparisons appear in Section A.10.

Changing arterial compliance has a different role. In these tests it changes the size of the pressure pulse more than the mean pressure. The distinction follows from the compartmental picture: arterial compliance determines how much blood can be stored during ejection, while the mean flow must still pass through the systemic resistance. These one-at-a-time tests illustrate mechanisms; they do not describe a patient population or establish which parameters could be recovered uniquely from measured data.

Recovery depends on how the inputs are restored as well as on the controller. We therefore repeat the raised-target experiment while delaying the restoration of resistance after the target has returned to its resting value. Some delays reduce the subsequent pressure peak, but the effect is not monotone: a delay equal to the pressure-filter time is not always the most effective choice. The timing of the next beat and the pressure already stored in the arterial compartment both matter. This suggests studying recovery as a coordinated change in the inputs, rather than assuming that one fixed delay will work under every condition.

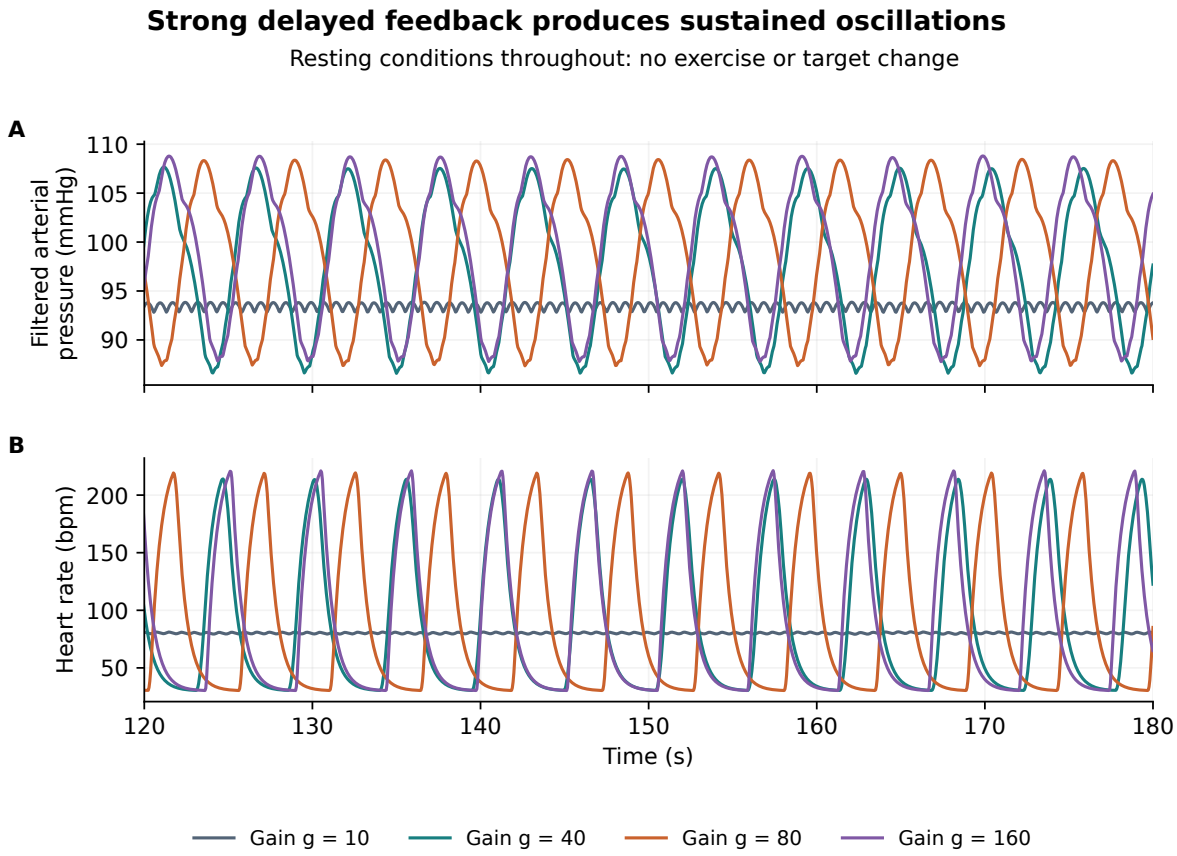


Figure 6: Longer simulations at resting resistance and a fixed target, with a 3 s pressure filter and a 0.6 s rate-response time. No exercise starts or stops in this experiment. The plots show the final minute of a three-minute run. Moderate gain produces a small repeating pressure variation; high gains produce much larger sustained oscillations in pressure and rate. The rate command is frequently bounded in the high-gain runs.

8 Discussion

The cardiac clock provides a simple way to change the rate of a pulsatile pump without changing the order of events within each beat. Combined with pressure feedback, it recovers a clear sequence of responses to vasodilation: pressure falls, the commanded rate rises, and increased flow offsets part of the pressure loss. A higher target produces a further rise in pressure and flow. These outcomes can be understood from the flow balances and the controller rather than from the plotted curves alone.

The same model makes the limits of this approach visible. Proportional feedback needs a pressure error to sustain an elevated rate. Higher gain reduces that error but makes the controller more sensitive to residual pressure pulses and response lag. Faster beating also does not preserve the volume ejected per beat. Finally, target changes can request rates outside a meaningful range unless bounds are included. Each limitation concerns a different part of the model: the feedback rule, the measurement and response times, or the pump's filling and ejection.

Several simplifications should guide interpretation. The inlet pressure is fixed, and the model omits venous return, the right heart, pulmonary circulation, wave propagation, and neural pathways.

Both systole and diastole shorten together as rate rises; there is no separate rate-dependent contractility law. The bounds and response times are illustrative choices rather than estimates from experimental data. The exercise inputs are sharp steps, making recovery sensitive to beat timing. These features make the model useful for examining mechanisms, but they limit direct physiological predictions.

The numerical checks support the internal consistency of the reported simulations. They do not turn the model into a validated representation of a particular person or exercise protocol. Likewise, the small parameter study is narrower than a global sensitivity analysis of a detailed regulated circulation [4]. Natural next steps include coupling the pump to a closed-volume venous compartment, testing smoother input changes at different cardiac phases, and comparing this proportional controller with strategies that can reduce the steady pressure error without excessive gain.

9 Conclusion

A continuously varying cardiac clock connects heart-rate feedback to a pulsatile circulation in a direct and physically interpretable way. The model shows how a faster heart can support pressure after vasodilation, why a fixed proportional controller leaves a pressure shortfall, and how raising the target changes both exercise and recovery. It also shows why bounds and response time matter: they keep the cardiac clock meaningful and determine whether pressure corrections settle or become sustained oscillations. The resulting framework offers a small, reproducible setting in which these mechanisms can be studied separately before combining them in a more complete circulation model.

A Advanced interpretation of the model

The preceding sections explain the circulation in terms of filling, ejection, pressure regulation, and recovery. Here we give the more formal interpretation of the same equations and deterministic experiments. The analysis separates four questions: whether the state remains physically admissible, what pressure error the controller must retain, whether small disturbances decay, and whether the numerical method resolves the dynamics. These questions require different arguments; success on one does not imply success on the others.

A.1 Cardiac phase as a change of time variable

The cardiac clock separates the prescribed contraction cycle from the rate at which it is traversed. Let $T_0 = 1/F_0$ and define $t_{\text{heart}} = T_0\phi$. Equation (6) then gives $\dot{t}_{\text{heart}} = F/F_0$. This reparameterizes the compliance waveform without changing its range. It uniformly shortens both systole and diastole as heart rate increases; it introduces no additional law for rate-dependent contractility or relaxation.

If $F > 0$, the successive integer crossings of ϕ define beat times t_k, with

$$\int_{t_k}^{t_{k+1}} F(s)\,ds = 1, \qquad \frac{1}{t_{k+1} - t_k} = \frac{1}{t_{k+1} - t_k}\int_{t_k}^{t_{k+1}} F(s)\,ds. \tag{16}$$

Thus reciprocal beat interval is the time average of the instantaneous phase speed over that beat. It need not equal the rate sampled at either endpoint. Strictly positive phase speed also permits phase to replace physical time when constructing the one-beat return map in Section A.7.

A.2 Positivity and admissible rate control

Proposition A.1 (Nonnegative stressed volumes)**.** *With positive resistances and compliances, nonnegative initial stressed volumes, and $P_{LA} \geq 0$, the nonnegative volume quadrant is forward invariant under the compartment equations.*

Proof. On $x_{\text{LV}} = 0$ with $x_{\text{A}} \geq 0$, ventricular pressure is zero and the aortic valve carries no flow. Hence $\dot{x}_{\text{LV}} = P_{\text{LA}}/R_{\text{Mi}} \geq 0$. On $x_{\text{A}} = 0$, systemic outflow vanishes and $\dot{x}_{\text{A}} = Q_{\text{Ao}} \geq 0$. The vector field therefore points into the nonnegative quadrant at either boundary. The positive-part valve laws are locally Lipschitz in the volumes, and the prescribed compliance is positive and continuous, so the boundary argument applies across its piecewise smooth cycle. □

Proposition A.2 (Admissible phase speed)**.** *For $g > 0$, the unrestricted proportional command is positive precisely when $p_f < P_*(1 + 1/g)$. If the bounded response starts with $F(0) \in [F_{\min}, F_{\max}]$ and $F_{\min} > 0$, that interval remains invariant and cardiac phase is strictly increasing.*

Proof. The first assertion follows by solving the command inequality $u > 0$. For $\tau_{\text{H}} > 0$, the bounded rate response has the representation

$$F(t) = e^{-t/\tau_{\text{H}}}F(0) + \frac{1}{\tau_{\text{H}}}\int_0^t e^{-(t-s)/\tau_{\text{H}}}u_b(s)\,ds. \tag{17}$$

Its nonnegative weights sum to one, so $F(t)$ stays between the same bounds as its initial value and the bounded command. For instantaneous response, $F = u_b$ gives the assertion directly. The positive lower bound then implies $\dot{\phi} > 0$. □

A downward target jump changes u immediately while filtered pressure remains continuous. At gain 10, resetting the target to approximately 93.47 mmHg requires $p_f < 102.82$ mmHg for the unrestricted command to remain positive. Delaying the resistance restoration does not guarantee this condition. Bounding the command prevents backward phase, but does not by itself eliminate a recovery overshoot or establish stability.

A.3 Exact mean regulation error

Proposition A.3 (Periodic proportional-control offset)**.** *Suppose the target and $g > 0$ are constant. On a periodic solution with inactive rate bounds, either instantaneous control or a first-order rate response satisfies Eq. (14). If systemic resistance is also constant, all three mean flows coincide as in Eq. (19) below.*

Proof. Averaging the filter over a period eliminates its derivative and gives $\overline{p_f} = \overline{P_{\mathrm{A}}}$. The rate equation similarly gives $\overline{F} = \overline{u}$ when a response lag is present. The command is affine in p_f, so

$$\overline{F} = F_0 \left[1 + g \frac{P_* - \overline{P_{\mathrm{A}}}}{P_*} \right]. \tag{18}$$

Rearrangement proves Eq. (14). Integrating each compartment's volume balance over a period also gives

$$\overline{Q_{\mathrm{Mi}}} = \overline{Q_{\mathrm{Ao}}} = \overline{Q_{\mathrm{s}}} = \frac{\overline{P_{\mathrm{A}}}}{R_{\mathrm{s}}}. \tag{19}$$

□

This is an exact proportional-control identity for the pulsatile system, rather than a constant-stroke-volume approximation. Maintaining a mean rate above F_0 requires mean pressure below the fixed target. For the fixed-target exercise experiment, the measured mean rate predicts an approximately 8.5 mmHg offset, consistent with the finite-window simulation. Raising the target changes the requested operating state; the size of that change remains a prescribed input. When saturation is active, the bounded command is no longer globally affine in filtered pressure, so the identity cannot be applied without an additional saturation term.

A.3.1 Finite-window averages

The periodic mean relation assumes that the filter and rate return to their starting values. Over an arbitrary interval of length L, their changes need not vanish. Integrating the filter and unsaturated rate response gives

$$\overline{P_{\mathrm{A}}} = P_* \left[1 + \frac{1 - (\overline{F} + \tau_{\mathrm{H}} \Delta F / L)/F_0}{g} \right] + \frac{\tau_{\mathrm{P}} \Delta p_f}{L}. \tag{20}$$

This formula applies for constant target, $g > 0$, and inactive bounds; use $\tau_{\mathrm{H}} = 0$ for the instantaneous rate rule. The endpoint terms explain why averages during recovery need not obey the simpler periodic relation. If the bounds are active, the command is no longer affine in filtered pressure and a saturation correction is also needed.

A.4 Operating curves and finite control authority

The fixed-rate experiments in Fig. 3 sample the relation among rate, mean pressure, and ejected volume per beat after transients have decayed. Each run uses a fixed resistance, 400 steps per actual beat, 60 beats of integration, and the last 10 beats for averaging. The decline in ejected volume per beat shows that increasing rate also changes filling and ejection. Mean output cannot be extrapolated by multiplying the resting ejected volume by an arbitrary rate.

At exercise resistance and a fixed rate of 240 bpm, the computed mean pressure is approximately 124.9 mmHg, below the 150 mmHg target. This is evidence of limited rate-control authority for the specified pump and reservoir. It is not a global upper bound over all time-varying controls. The operating curves concern equilibrated fixed-rate experiments; the controlled transients have additional dependence on phase, stored arterial volume, and controller memory.

A.5 Gain, filtering, and saturation as distinct effects

The 48-point gain/filter study in Fig. 4 separates mean tracking performance from late rate variation. Its first measure is the mean absolute filtered-pressure error over the exercise interval; its second is the range of heart rate over the final nine seconds of exercise. These are deterministic design diagnostics with different meanings.

Increasing gain reduces proportional offset in the unsaturated periodic relation but also amplifies the rate response to pressure pulses that remain after filtering. At gain 80 and a 0.375 s filter, the instantaneous bounded rate spans 30–240 bpm and reaches a bound for approximately 80% of the exercise interval. A longer filter attenuates beat-scale variation while slowing the response to a change in mean pressure. Consequently, neither a small filtered error nor a large late rate range alone is a stability classification. The former may obscure a transient; the latter includes modulation at the cardiac period. Local orbital stability requires the return-map calculation below.

A.6 Averaged dynamics and the effect of rate-response lag

To understand this mechanism, we first average out the individual pressure pulses. Let $\mathcal{Q}(F, P)$ be the mean ejection of a ventricle beating at constant rate F against a fixed arterial pressure P. The corresponding approximate pressure equation is

$$C_{\mathrm{A}}\dot{P} = \mathcal{Q}(F, P) - P/R_{\mathrm{s}}. \tag{21}$$

This has the same inflow-minus-outflow structure as the full arterial compartment, but uses an averaged pump inflow. It is useful for interpreting slow changes and is not an exact replacement for the pulsatile model.

Near an operating state, write small changes in pressure, filtered pressure, and rate as p, z, and f. The linearized equations are

$$\dot{p} = -ap + bf, \qquad \tau_{\mathrm{P}}\dot{z} = p - z, \qquad \tau_{\mathrm{H}}\dot{f} = -Kz - f, \tag{22}$$

where

$$a = \frac{R_{\mathrm{s}}^{-1} - \mathcal{Q}_P}{C_{\mathrm{A}}}, \qquad b = \frac{\mathcal{Q}_F}{C_{\mathrm{A}}}, \qquad K = \frac{F_0 g}{P_*}. \tag{23}$$

Each coefficient has a direct interpretation. The term a describes how pressure relaxes through outflow and the pump's response to pressure. The term b measures how strongly a rate increase raises pump inflow. The term K is the controller's response to a pressure error.

Without a rate-response lag, Eq. (22) has characteristic polynomial

$$\lambda^2 + (a + \tau_{\mathrm{P}}^{-1})\lambda + (a + bK)/\tau_{\mathrm{P}}. \tag{24}$$

Its roots have negative real parts when $a > 0$ and $bK \geq 0$. With a positive lag the polynomial becomes

$$(\lambda + a)(\tau_{\mathrm{P}}\lambda + 1)(\tau_{\mathrm{H}}\lambda + 1) + bK. \tag{25}$$

For positive a, b, K, the cubic Routh–Hurwitz condition is

$$[\tau_{\mathrm{P}} + \tau_{\mathrm{H}} + a\tau_{\mathrm{P}}\tau_{\mathrm{H}}][1 + a(\tau_{\mathrm{P}} + \tau_{\mathrm{H}})] > \tau_{\mathrm{P}}\tau_{\mathrm{H}}(a + bK). \tag{26}$$

The right side increases with controller gain. Thus even though each part of the loop responds in the intended direction, the combined response can lose stability when the gain becomes too large.

For the clamped-afterload pump at 80 bpm, the reduced equilibrium pressure is 93.9384 mmHg. Central differences with perturbations of 1 bpm and 0.5 mmHg give $\mathcal{Q}_F \simeq 0.06192$ L/beat and $\mathcal{Q}_P \simeq -0.002400$ L/(min mmHg). The pump calculation uses 800 steps per cycle over 12 cycles. These values give $a \simeq 33.37$ min^{-1} and $b \simeq 35.38$ mmHg/(min bpm). With a 3 s filter and a 0.6 s rate response, Eq. (26) gives a critical gain near 14.17. This approximation uses its own equilibrium pressure as the target; the full pulsatile model retains the target in Table 1.

A.7 Return-map and Floquet interpretation of the pulsatile solution

Let $\mathcal{P}$ take a state on $\phi = 0$ to its next crossing at $\phi = 1$. A repeating beat is a fixed point y_* with $\mathcal{P}(y_*) = y_*$. To first order, a transverse perturbation evolves as

$$\delta y_{k+1} = D\mathcal{P}(y_*)\, \delta y_k. \tag{27}$$

The eigenvalues of $D\mathcal{P}(y_*)$ approximate the transverse Floquet multipliers. A spectral radius below one implies local decay of sufficiently small perturbations; a multiplier outside the unit circle indicates local instability. This is an asymptotic classification, not a claim that every disturbance decreases in norm on every beat.

The full-model return map is defined from phase zero to phase one using ventricular pressure, arterial pressure, filtered pressure, and rate as its state. Fixing phase removes the neutral shift in the timing of the beat. Periodic states are found by solving for a fixed point of this map. Centered finite differences of 10^{-3} and 10^{-4} in each state coordinate check the map derivative. At 1,600 steps per beat, the periodic-root residuals are below 2×10^{-11} in the corresponding state units. The commanded rate stays strictly between its bounds on these orbits. An independent phase-

Gain	400 steps	800 steps	1,600 steps	Independent Radau
10	0.93957	0.94016	0.93984	–
12	0.96690	0.96762	0.96728	–
14	0.99227	0.99310	0.99275	0.99292
16	1.01607	1.01700	1.01665	1.01685
20	1.05998	1.06112	1.06074	–
40	1.23659	1.23861	1.23806	–

Table 3: Largest one-beat multiplier magnitude for the resting orbit with gradual rate response. The classification changes from stable to unstable between gains 14 and 16.

based Radau integration gives magnitudes 0.99292 at gain 14 and 1.01685 at gain 16. Derivative

perturbations of 0.01 and 0.003 agree within 2×10^{-7}. Without rate-response lag, the corresponding 1,600-step calculation gives 0.71763 at gain 10 and 0.71321 at gain 40. These tests support the reported stability classification; they do not locate an exact critical gain or establish a nonlinear bifurcation theorem.

A.8 Local instability and sustained bounded oscillations

The resting time simulations in Fig. 6 address the nonlinear response after startup. They use a 3 s pressure filter and a 0.6 s rate-response time, with neither an exercise step nor a target change. Over the last 30 seconds of each three-minute run, the filtered-pressure range is approximately 1.05 mmHg at gain 10 and 21 mmHg at gains 40, 80, and 160. The bounded rate command reaches its limits for approximately 79%, 91%, and 95% of that window in the three high-gain cases.

These observations complement the return map. The multiplier calculation concerns infinitesimal disturbances near a one-beat orbit with inactive bounds, whereas the time simulations show finite-amplitude responses in which saturation is frequent. With a finite response time, the actual rate approaches the command gradually and need not itself reach the bounds when the command does. The numerical traces demonstrate persistent bounded oscillations over the simulated interval; they do not prove uniqueness of an attractor, identify a bifurcation type, or establish global stability.

A.9 Numerical consistency and time resolution

A.9.1 A pressure update that respects the flow balances

At each time step of length h, the controller and cardiac phase are advanced, and the two compartment pressures are then obtained from a backward-Euler update. If the valve states are fixed temporarily, put $g_m = S_m/R_{\mathrm{Mi}}$ and $g_a = S_a/R_{\mathrm{Ao}}$, with $S_m, S_a \in \{0, 1\}$. The pressure system is

$$\begin{pmatrix} C^{n+1} + h(g_m + g_a) & -hg_a \\ -hg_a & C_{\mathrm{A}} + h(g_a + R_{\mathrm{s}}^{-1}) \end{pmatrix} \begin{pmatrix} P_{\mathrm{LV}}^{n+1} \\ P_{\mathrm{A}}^{n+1} \end{pmatrix} = \begin{pmatrix} C^n P_{\mathrm{LV}}^n + hg_m P_{\mathrm{LA}} \\ C_{\mathrm{A}} P_{\mathrm{A}}^n \end{pmatrix}. \tag{28}$$

We test the four valve configurations and accept the one consistent with the calculated pressure differences. This small enumeration avoids repeatedly changing valve states without a termination criterion.

The update has two useful properties. First, the determinant is positive: the product of the diagonal entries contains $(hg_a)^2$ plus positive terms. Its inverse has nonnegative entries, so nonnegative initial pressures and inlet pressure give nonnegative updated pressures. Second, adding the two rows gives

$$\frac{x_{\mathrm{LV}}^{n+1} - x_{\mathrm{LV}}^n + x_{\mathrm{A}}^{n+1} - x_{\mathrm{A}}^n}{h} = Q_{\mathrm{Mi}}^{n+1} - Q_{\mathrm{s}}^{n+1}. \tag{29}$$

The volume gained by the model equals the amount entering minus the amount leaving. This is the balance for the open system in Fig. 1, not conservation of a fixed total blood volume.

The pressure solution is unique. One way to see this is that it minimizes the strictly convex function

$$\begin{aligned} \mathcal{J}(v, a) = {} & \tfrac{1}{2} C^{n+1} v^2 + \tfrac{1}{2} C_{\mathrm{A}} a^2 - C^n P_{\mathrm{LV}}^n v - C_{\mathrm{A}} P_{\mathrm{A}}^n a \\ & + \frac{h}{2R_{\mathrm{Mi}}} [P_{\mathrm{LA}} - v]_+^2 + \frac{h}{2R_{\mathrm{Ao}}} [v - a]_+^2 + \frac{h}{2R_{\mathrm{s}}} a^2. \end{aligned} \tag{30}$$

Differentiating gives the implicit flow equations. The positive quadratic storage terms ensure a unique minimum. At a zero valve pressure difference, either valve-state convention gives zero flow.

The pressure filter uses the held-input update

$$p_f^{n+1} = p_f^n + (1 - e^{-h/\tau_P})(P_A^n - p_f^n). \tag{31}$$

The rate response uses the analogous formula with a held command, and phase advances by hF^{n+1}. The full coupled method is first order. Input changes are applied at the new time index. A phase-based calculation for the one-beat map instead uses a fixed phase increment and physical time increment $1/(NF)$, where N is the number of steps per beat.

A.9.2 Time-step refinement and an independent solver

The calculations were repeated at progressively smaller steps without retuning the pressure target. Table 4 shows the raised-target experiment with gradual rate response. Coarse time resolution can change whether the upper rate bound is reached, so a small balance residual alone is not evidence of temporal accuracy. At 400 steps per baseline cycle, mean pressure differs from the 1,600-step

Steps per baseline cycle	Mean P_A mmHg	Mean Q_s L/min	Mean F bpm
100	117.466	13.0518	240.000
200	120.509	13.3899	237.460
400	121.120	13.4578	234.169
800	121.407	13.4897	232.536
1600	121.495	13.4995	231.698

Table 4: Step refinement for the raised-target experiment with a 0.6 s rate response; means use 51–60 s.

result by about 0.31% and mean rate by about 1.07%. The two finest mean pressures differ by about 0.07%. In a separate fixed-rate comparison over 12–18 s, an adaptive Radau integration in stressed-volume variables gives mean pressure 93.7833 mmHg, compared with 93.6424 mmHg for the 1,600-step method, a difference of 0.15%.

The maximum combined balance residual in the four main traces is below 5.3×10^{-12} L/min. In 1,000 randomized pressure updates, both compartment balances, the valve inequalities, and nonnegative pressures and flows were checked; the largest combined residual was 2.6×10^{-10} L/min. The random test seed is recorded in the accompanying code. These checks verify the implementation of the model equations.

All plotted experiments are deterministic. Fixed-rate curves use 400 steps per actual cycle, integrate for 60 cycles, and average over the last 10. The gain/filter grid uses gains $0, 1, 2, 5, 10, 20, 40, 80$ and filter times $0.375, 0.75, 1.5, 3, 6, 12$ s. The full equations, parameters, Python implementation, saved traces, and scripts for reproducing the figures and tables accompany the manuscript.

A.10 Parameter perturbations and recovery scheduling

Changed parameter	Multiplier	Mean pressure (mmHg)	Mean rate (bpm)
P_{LA}	0.8	80.94	187.50
P_{LA}	1.2	87.71	128.31
C_{A}	0.8	84.98	152.73
C_{A}	1.2	85.00	152.27
C_{D}	0.8	81.53	183.12
C_{D}	1.2	87.62	130.80
R_{Mi}	0.8	85.40	149.86
R_{Mi}	1.2	84.61	155.24
R_{Ao}	0.8	84.98	152.58
R_{Ao}	1.2	84.98	152.58

Table 5: Changing one parameter at a time while retaining fixed-target feedback, gain 10, a 3 s pressure filter, and instantaneous bounded rate response. Averages use 51–60 s.

For the recovery comparison, the pressure target returns to rest at 60 s, while restoration of resistance is delayed by 0, 0.75, 1.5, 3, or 6 s. With a 0.6 s rate response, the corresponding raw pressure peaks in the first 15 s of recovery are 143.3, 128.6, 131.7, 146.6, and 137.3 mmHg. This comparison uses one event phase and the main time step. The nonmonotone pattern motivates a broader study of event timing rather than selecting a universal delay from these five cases.

A.11 What the combined analysis establishes

The phase construction, positivity arguments, mean-error relation, operating curves, and stability tests answer complementary questions. Positive phase speed preserves the order of cardiac events. Nonnegative volumes and the discrete balance preserve the compartment interpretation. The mean identity explains the fixed-target pressure deficit without replacing the pulsatile pump by a constant stroke volume. The operating curves describe the pump's rate-dependent delivery under fixed conditions. Finally, the averaged criterion explains a mechanism for overcorrection, and the full-model return map tests that mechanism in the beating circulation.

The finite time-step and independent-solver checks support the reported numerical results. They do not constitute a physiological calibration or a proof of the nonlinear long-time dynamics. In particular, the gain/filter grid, the local stability calculation, and the sustained-oscillation runs should be read together while retaining their different assumptions and measurement windows.

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